MAIS PAC

IPhO 03.03: Data Analysis and Linear Fitting · Lecture + Problems

Data Analysis and Linear Fitting

MAIS Physics & Astronomy Club (PAC) · IPhO lecture + problem handout

6 lecture sections · 10 problems · 1 interactive experiment · 5 vector figures

Start with a physical picture, then introduce only the mathematics needed to make it predictive. The lecture connects everyday reasoning to olympiad measurements. Five vector diagrams and an interactive experiment support the argument; ten graded problems and independent worked solutions turn the ideas into practice. The coverage checklist is the IPhO syllabus, not this handout.

Contents & learning path

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1. From numbers to physics

A table of measurements is not a result. The physics comes out of the **pattern** in the table: a gradient, an intercept, a curve that flattens, a difference that grows linearly. Data analysis is the discipline of extracting that pattern and saying honestly how well you know it.

Three operations cover almost everything you will be asked to do:

- **Linearise** the theory, so that the pattern is a straight line.
- **Fit** the line, using all the points and their uncertainties.
- **Propagate** the uncertainty of the fit into the physical quantity you actually want.

2. Linearisation

Almost every school experiment reduces to y = mx + c. The trick is to plot the combination of measured quantities that theory says is linear.

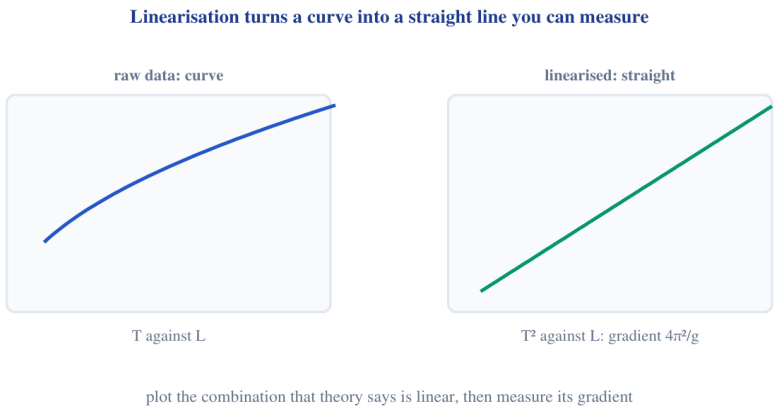


Figure 1. A gradient can be measured with a ruler and compared with theory; a curvature cannot. Linearising is the single most valuable data-handling skill in the practical exam.

experiment	plot	gradient	intercept
pendulum	T² vs L	4π²/g	should be 0
lens	1/v vs 1/u	-1	1/f
Ohm's law with a real cell	V vs I	-r	ε
cooling	ln (T-T_0) vs t	-k	ln (T_({start})-T_0)
absorption	ln I vs x	-μ	ln I_0
gas law	p vs 1/V	nRT	0

For a power law y = ax^n, take logarithms:

$$\log y = \log a + n\log x, \tag{1}$$

so a log–log plot has gradient n and intercept log a. For an exponential y = y_0e^(-kx), a semi-log plot (log y against x) is a straight line of gradient -k.

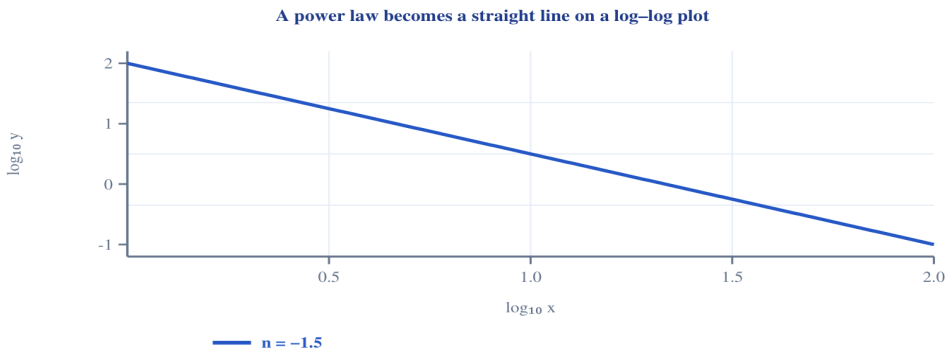


Figure. If y = a·x■ then log y = log a + n log x: the gradient is the exponent n and the intercept is log a.

3. The line of best fit

By eye, the line should pass through the "middle" of the scatter with roughly equal numbers of points above and below, and it must not be forced through the origin unless theory demands it.

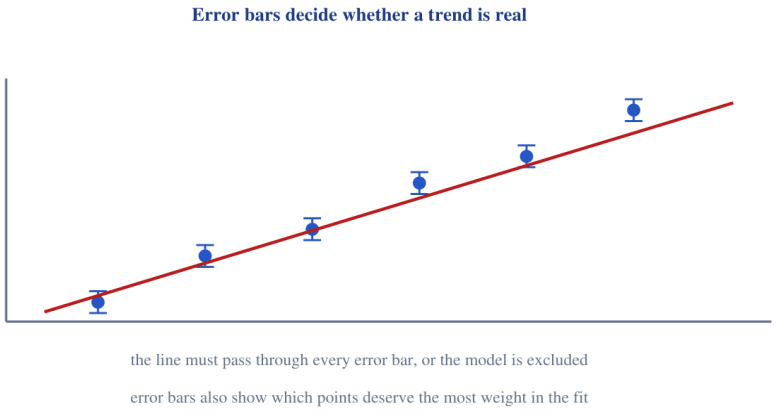


Figure 4. Without error bars there is no way to tell a real deviation from scatter. Drawing them forces you to think about the uncertainty of every point, which is where most of the physics of a practical question actually lies.

Numerically, the **least-squares** fit minimises $\Sigma (y_i - mx_i - c)^2$, which gives

$$m = (\Sigma (x_i - x)(y_i - y))/(\Sigma (x_i - x)^2), \quad c = y - m x. \tag{2}$$

The uncertainty in the gradient is

$$\Delta m = \sqrt{((\Sigma (y_i - mx_i - c)^2)/((N-2)\Sigma(x_i - x)^2))}, \tag{3}$$

which grows with the scatter and shrinks with the spread of x — the mathematical statement of "span the range".

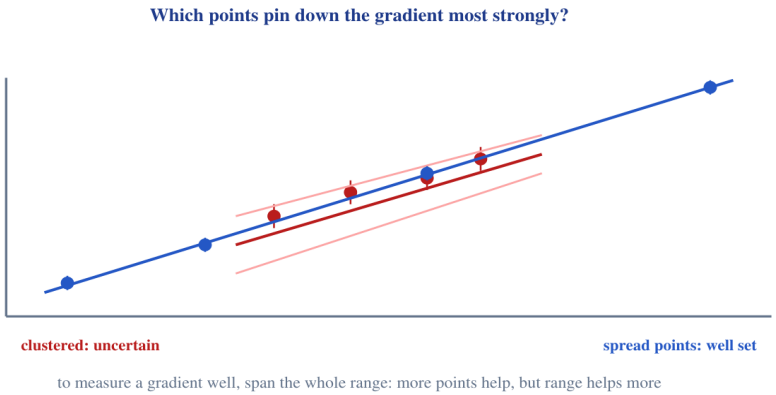


Figure 3. This is why an experiment sweeps the full range: ten points in a narrow interval determine a gradient worse than three points spread across the whole scale.

The **coefficient of determination** $R^2 = 1 - {SS}_({res})/{SS}_({tot})$ measures how much of the variation the line explains. $R^2 = 1$ is a perfect line; above 0.99 is excellent for experimental data; below 0.9 suggests either large scatter or the wrong model. R^2 is not a substitute for plotting: a curved set of points can have a high R^2 and a completely wrong gradient.

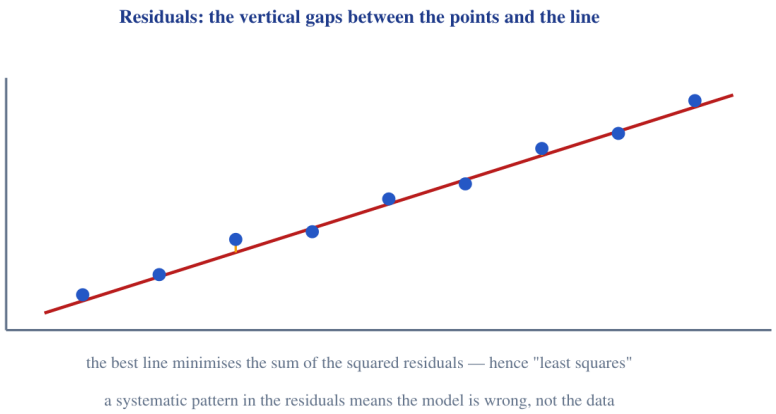


Figure 2. If the residuals wander systematically — all positive at one end, all negative at the other — a straight line is the wrong model, or there is a systematic error in one of the variables.

Interactive experiment: Least-squares fitting — open the online handout to use it.

4. Residuals: how to tell a model is wrong

A residual is y_i - (mx_i+c). If the fit is right and the errors are random, the residuals should wander above and below zero with no pattern. If they show a systematic shape — all positive at one end, all negative at the other — then either the model is wrong (the relationship is curved, not linear) or one variable has a systematic error.

That single diagnostic is often the highest-scoring part of a practical question, because it shows you understand that fitting is a **test** of the model, not just an exercise in drawing.

5. Reporting the analysis

For a graph question, the expected answer has a shape:

1. State what is plotted against what, and why that linearises the theory.
2. Quote the gradient (and intercept) with units and an uncertainty obtained from the extreme acceptable lines.
3. Convert the gradient into the physical quantity, propagating the uncertainty.
4. Compare with the accepted value, in units of the uncertainty, and comment on any discrepancy.
5. Mention at least one source of systematic error and how it would shift the gradient or intercept.

The data-analysis checklist. Linearise before you plot: write the theory as y = mx + c and identify y, x, m and c → plot with error bars, label axes with units → draw the best line by eye or fit it by least squares → get the uncertainty from the steepest and shallowest acceptable lines → convert the gradient into the wanted quantity and propagate its uncertainty → check the residuals for a pattern → compare with the accepted value, and explain any discrepancy physically rather than apologising for it.

6. Problems: reason, calculate, check

Ten original problems in three tiers: **[1]** foundation, **[2]** multi-step olympiad reasoning, **[3]** challenge. Use the constants supplied in the lecture unless a question overrides them. State assumptions, keep units, and estimate the answer before computing it. Every problem has a worked solution with a consistency check.

[1] Problem 1 — Linearising a spring. For a mass on a spring, theory gives $T = 2\pi\sqrt{(m/k)}$. (a) State what should be plotted against what to give a straight line, and what the gradient represents. (b) A graph of T^2 against m has a gradient of 0.82 {s²,kg^(-1)}. Find k. (c) Explain why the intercept of such a graph is often slightly positive in a real experiment.

Worked solution →

[1] Problem 2 — A log–log plot. Data for a falling object give s against t. (a) Show that $s = (1)/(2)at^2$ is a power law and state the exponent. (b) Find the gradient expected on a log–log plot. (c) Explain how the intercept gives a.

Worked solution →

[2] Problem 3 — Least squares by hand. Four points are (1.0, 2.1), (2.0, 3.9), (3.0, 6.2), (4.0, 7.8). (a) Find the means of x and y. (b) Find the least-squares gradient. (c) Find the intercept and comment on whether it is consistent with zero.

Worked solution →

[2] Problem 4 — Uncertainty in a gradient. For the data of Problem 3, the sum of squared residuals is 0.036. (a) Estimate the uncertainty in the gradient. (b) Find the percentage uncertainty. (c) Explain which change to the experiment would reduce it most: more points, or the same number spread over a wider range.

Worked solution →

[2] Problem 5 — Residuals as a diagnostic. A student plots V against I for a filament lamp and gets residuals that are positive at both ends and negative in the middle. (a) Sketch the shape of the residuals and explain what it implies. (b) State what the student should plot instead if the resistance increases with temperature. (c) Explain why a straight line is still useful for a small range of currents.

Worked solution →

[2] Problem 6 — Interpreting R^2. Three fits give $R^2 = 0.9995$, 0.998 and 0.87. (a) Rank them by quality and explain what each value suggests. (b) Give an example of a dataset with high R^2 but the wrong model. (c) Explain why R^2 alone is not enough to justify a fit.

Worked solution →

[2] Problem 7 — Cooling data. A cup of water cools in a room at 20°{C}; the excess temperature falls from 60 {K} to 30 {K} in 8.0 {min}. (a) Show that a plot of ln (T-20) against t should be a straight line and state its gradient. (b) Find the cooling constant from the two values given. (c) Find the excess temperature after 20 {min} and explain the role of the intercept.

Worked solution →

[3] Problem 8 — Two variables with uncertainties. In a pendulum experiment both L and T have uncertainties. (a) Explain why the uncertainty in T usually dominates the uncertainty in g. (b) Find the ratio of the contributions for L = (1.000 ± 0.002) {m} and T = (2.006 ± 0.004) {s}. (c) Suggest how a student could balance the two contributions.

Worked solution →

[3] Problem 9 — Weighted fitting. Two measurements of the same quantity give 5.0 ± 1.0 and 8.0 ± 0.5. (a) Find the simple mean. (b) Find the weighted mean, using weights $1/\sigma^2$. (c) Explain why the weighted mean lies closer to the more precise value, and find its uncertainty.

Worked solution →

[3] Problem 10 — A complete analysis. A student measures the resistance of a wire at several lengths and obtains: (0.20 {m}, 0.86 Ω), (0.40, 1.70), (0.60, 2.55), (0.80, 3.42), (1.00, 4.26), with an uncertainty of ±0.03 Ω in each resistance. (a) Plot or tabulate what you would graph and find the gradient. (b) Find the resistivity given a diameter of (0.40 ± 0.01) {mm}. (c) Find the uncertainty in the resistivity from the gradient, and state the dominant contribution.

Worked solution →

Further practice and provenance

The 10 exercises above and their solutions are **original MAIS PAC problems**, inspired by the skills in this IPhO syllabus topic. They are not copied from official papers and are not official marking schemes.

Sources and scope. Topic coverage follows the IPhO syllabus (accepted 2014, amended 2015) as published by the IPhO organisation. Official problems are available through the IPhO past-paper archive. The derivations, diagrams, experiments and numerical teaching examples in this handout were prepared independently for MAIS PAC. Calculator-level methods are emphasized; optional calculus connections are explicitly identified. Synthetic data, when used, are labelled as teaching data.

Where to learn more

IPhO syllabus and statutes

The coverage checklist this curriculum follows: accepted 2014, amended 2015. Read it once for scope, not for depth.

IPhO past problems and solutions

The only honest test of readiness. Work recent theoretical and experimental papers under time, then compare with the official marking schemes.

BIPM: SI brochure and uncertainty guidance

The GUM is the reference behind the uncertainty rules used in 03.02 and 03.03.

Particle Data Group: mathematical tools review

A compact professional summary of means, errors, fits and confidence intervals.